Aditya Meenakshisundaram

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Education

Texas A&M University May 2028
Bachelor of Science in Chemical Engineering
Cumulative GPA: 3.86 / 4.00 | Craig & Galen Brown Engineering Honors
Awards: Samsung Semiconductor Fellowship, TEX-E Fellowship, AWS Scholarship

Work Experience & Research

Photolithography & Metrology Intern May 2026 – Present
Samsung Austin Semiconductor Austin, TX

- Cut test-wafer qualification time by 15% through a predictive photoresist modeling application optimizing spin coater dispense parameters within a 10-Å film thickness target & automating equipment adjustments
- Protected downstream device yield by engineering an overlay correction algorithm transforming 500+ wafer- & shot-level measurements into lot-level scanner offsets, surfacing abnormal multi-layer alignment behavior at source
- Built Python, SQL, & Streamlit data pipelines parsing & validating 5,000+ metrology data points across 20+ scanner & spinner toolsets, removing manual process control overhead
- Developed a Convolutional Neural Network (CNN) based defect classification system ingesting continuous lithography tool image streams to trigger real-time technician alerts, targeting elimination of delayed-detection rework cycles

Semiconductor Fellow Jan 2026 – Present
Samsung Austin Semiconductor College Station, TX

- Demonstrated passive thermal-switching behavior in 1–2 mm indium/titanium nanowire composite formulations designed to suppress heat propagation during lithium-ion battery thermal runaway events
- Mapped composition-to-switching-threshold relationships by resolving thermal conductivity response across 50 wt% & 75 wt% nanowire-loaded formulations, varying porosity, composition, & indium content
- Traced thermal-switching thresholds & mechanical integrity to microstructure & conductive pathway formation via SEM cross-section analysis across 4+ formulation iterations, directly guiding reformulation decisions

Silicon Carbide (SiC) Researcher Aug 2025 – Present
Green Group, Texas A&M University College Station, TX

- Established a reproducible polycarbosilane-to-SiC conversion processing window across 10+ pyrolysis iterations (800–1200°C), narrowing ramp rate & atmosphere parameters to viable conditions for high-power semiconductor integration
- Isolated cracking mechanisms & identified variables controlling conversion quality by linking thermal processing conditions to SiC film morphology, ceramic yield, & conductivity via SEM, TGA, & four-point probe
- Demonstrated RF-driven volumetric heating as a faster alternative to conventional furnace pyrolysis for SiC fiber architectures, improving heating uniformity across fiber cross-sections

Projects

Aggie Research Finder Co-Creator; Lead Developer

- Reached 3,000+ active student users by building & deploying a centralized research matching platform indexing 1,770+ Texas A&M faculty members across 15+ departments
- Eliminated manual faculty data curation by building a custom Python scraper automating extraction & continuous updates of research interests & lab affiliations from 15+ distributed institutional sources

Sustinapath Process Analytics Platform

- Quantified chemical process sustainability & economics end-to-end by generating 0–100 composite scores across time, cost, & sustainability dimensions using custom thermodynamic & economic scoring algorithms in Python
- Replaced manual benchmarking workflows with an automated reporting pipeline surfacing process bottlenecks & delivering targeted optimization recommendations

Substack Advanced Materials, Energy Storage & Semiconductor Technology

- Grew to 400+ monthly readers; 20+ articles on semiconductor technology, advanced materials, & energy storage
- Covered EUV lithography, advanced packaging, & supply chain dynamics with engineering grounded analysis

Technical Skills

Materials & Processes: Semiconductor Engineering, Pyrolysis, Nanowire Fabrication, Thermodynamics, Thin Film Processing
Characterization: SEM, XRD, FTIR, TGA, Optical Microscopy, Four-Point Probe, Thermal Conductivity Measurement
Programming & Data: Python (NumPy, Pandas, scikit-learn, pytorch), SQL, Streamlit, HTML, Git/Version Control